

Week6 day1 haystack

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Command Prompt

(venv) C:\Users\DELL\Downloads\week6_day1_haystack>python pipeline.py
PDF files found: 5
Documents loaded: 5
Documents indexed: 5

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Question 1: What is artificial intelligence?
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Result 1
Score: 1.9180818684818406
Content: Artificial Intelligence
Artificial Intelligence (AI) is a branch of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, perception, and understanding language. AI is used in recommendation systems, virtual assistants, healthcare, finance, robotics, and autonomous vehicles.

Result 2
Score: 1.4842603860123846
Content: Natural Language Processing
Natural Language Processing (NLP) is an area of artificial intelligence that enables computers to process and understand human language. NLP applications include sentiment analysis, machine translation, text classification, chatbots, speech recognition, and question answering. Modern NLP systems often use transformer-based models.

Result 3
Score: 1.4532713430009907
Content: Machine Learning
Machine Learning (ML) is a subset of artificial intelligence in which computers learn patterns from data instead of being explicitly programmed for every task. The main types are supervised learning, unsupervised learning, and reinforcement learning. Common applications include spam detection, prediction, classification, fraud detection, and recommendation systems.

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Question 2: What is machine learning?
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Result 1
Score: 2.0066408065310366
Content: Machine Learning
Machine Learning (ML) is a subset of artificial intelligence in which computers learn patterns from data instead of being explicitly programmed for every task. The main types are supervised learning, unsupervised learning, and reinforcement learning. Common applications include spam detection, prediction, classification, fraud detection, and recommendation systems.

Result 2
Score: 1.775484046470769
Content: Deep Learning
Deep Learning is a part of machine learning that uses neural networks with multiple layers. Deep learning models can automatically learn useful features from large amounts of data. Convolutional neural networks are commonly used for images, while recurrent and transformer-based architectures are used for sequential and language tasks.

Result 3
Score: 1.1449420947807691
Content: Artificial Intelligence
Artificial Intelligence (AI) is a branch of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, perception, and understanding language. AI is used in recommendation systems, virtual assistants, healthcare, finance, robotics, and autonomous vehicles.

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Question 3: What is supervised learning?
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Result 1
Score: 2.895330564217541
Content: Machine Learning
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Question 4: What is cybersecurity?
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Result 1
Score: 2.2706277569426396
Content: Cybersecurity
Cybersecurity is the practice of protecting computers, networks, applications, and data from unauthorized access, attacks, damage, or disruption. Important principles include confidentiality, integrity, and availability. Common threats include phishing, malware, ransomware, denial-of-service attacks, and social engineering. Security controls include authentication, encryption, firewalls, monitoring, and regular updates.

Result 2
Score: 1.0021382578895273
Content: Artificial Intelligence
Artificial Intelligence (AI) is a branch of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, perception, and understanding language. AI is used in recommendation systems, virtual assistants, healthcare, finance, robotics, and autonomous vehicles.

Result 3
Score: 0.9772898165761426
Content: Natural Language Processing
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Question 5: What is cloud computing?
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Result 1
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Question 5: What is cloud computing?
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Result 1
Score: 0.13570428218959554
Content: Artificial Intelligence
Artificial Intelligence (AI) is a branch of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, perception, and understanding language. AI is used in recommendation systems, virtual assistants, healthcare, finance, robotics, and autonomous vehicles.

Result 2
Score: 0.1108558408762109
Content: Natural Language Processing
Natural Language Processing (NLP) is an area of artificial intelligence that enables computers to process and understand human language. NLP applications include sentiment analysis, machine translation, text classification, chatbots, speech recognition, and question answering. Modern NLP systems often use transformer-based models.

Result 3
Score: 0.10854134373450221
Content: Machine Learning
Machine Learning (ML) is a subset of artificial intelligence in which computers learn patterns from data instead of being explicitly programmed for every task. The main types are supervised learning, unsupervised learning, and reinforcement learning. Common applications include spam detection, prediction, classification, fraud detection, and recommendation systems.
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Command Prompt
prediction, classification, fraud detection, and recommendation systems.

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Question 6: What are the advantages of cloud computing?
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Result 1
Score: 2.319891142288721
Content: Machine Learning
Machine Learning (ML) is a subset of artificial intelligence in which computers learn patterns from data instead of being explicitly programmed for every task. The main types are supervised learning, unsupervised learning, and reinforcement learning. Common applications include spam detection, prediction, classification, fraud detection, and recommendation systems.

Result 2
Score: 2.034637228468006
Content: Deep Learning
Deep Learning is a part of machine learning that uses neural networks with multiple layers. Deep learning models can automatically learn useful features from large amounts of data. Convolutional neural networks are commonly used for images, while recurrent and transformer-based architectures are used for sequential and language tasks.

Result 3
Score: 1.7749656730853456
Content: Cybersecurity
Cybersecurity is the practice of protecting computers, networks, applications, and data from unauthorized access, attacks, damage, or disruption. Important principles include confidentiality, integrity, and availability. Common threats include phishing, malware, ransomware, denial-of-service attacks, and social engineering. Security controls include authentication, encryption, firewalls, monitoring, and regular updates.

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Question 7: What is a neural network?
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